

Multiple Choice (10 marks)

1. Kirkwood gaps are observed in the main asteroid belt including at the position(s) where:
 - (a) asteroids would orbit with a period half that of Jupiter
 - (b) asteroids would orbit with a period twice that of Jupiter
 - (c) asteroids would orbit with a period twice that of Mars
 - (d) A and B
2. What are the conditions necessary for a terrestrial planet to have a strong magnetic field?
 - (a) fast rotation only
 - (b) a rocky mantle only
 - (c) a molten metallic core only
 - (d) both a molten metallic core and reasonably fast rotation
3. We were first able to accurately measure the diameter of Pluto from:
 - (a) a New Horizons flyby in the 1990 s
 - (b) Hubble Space Telescope images that resolved Pluto's disk
 - (c) brightness measurements made during mutual eclipses of Pluto and Charon
 - (d) radar observations made by the Arecibo telescope
4. Which is not a similarity between Saturn and Jupiter's atmospheres?
 - (a) a composition dominated by hydrogen and helium
 - (b) the presence of belts zones and storms
 - (c) an equatorial wind speed of more than 900 miles per hour
 - (d) significant "shear" between bands of circulation at different latitudes
5. The terrestrial planet cores contain mostly metal because
 - (a) the entire planets are made mostly of metal.
 - (b) metals condensed first in the solar nebula and the rocks then accreted around them.
 - (c) metals sank to the center during a time when the interiors were molten throughout.
 - (d) radioactivity created metals in the core from the decay of uranium.

True / False (10 marks)

Answer True or False

6. True or False: The number of sunspots peaks every 9 years, which is referred to as the solar cycle.
7. True or False: The lithosphere is primarily composed of the softer rocky material found in the mantle beneath the crust.
8. True or False: The visible spectrum of a star viewed through cool hydrogen gas exhibits absorption lines superimposed on the star's blackbody spectrum.
9. True or False: Impact heating is a significant mechanism for maintaining a stable atmosphere on Mercury despite its proximity to the Sun.
10. True or False: Iron is the second most common element in the solar system, ranking just after hydrogen.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between mass, density, and size in gas giants, specifically explaining why Saturn, despite its smaller mass, is nearly as large as Jupiter.
12. Discuss the arrangement of electromagnetic radiation categories from shortest to longest wavelength, analyzing their significance in astronomical observations and applications.

End of Paper